

# Harsath Kaliappa Gounder Thangavel

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## Education

### York University

Bachelor of Science (BSc) in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

## Professional Experience

### Software Developer

Privex Inc (privex.io)

October 2023 – January 2024

London, United Kingdom (remote)

- Developed an AI-driven customer support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, delivering personalized responses to 800+ customers about support tickets, products, and services.
- Reduced support staff workload by 80% while providing instantaneous, context-aware assistance that improved overall customer experience.
- Implemented REST APIs to interface with the LLM and a Redis-based asynchronous task queue for processing customer queries and staff updates via Discord, prioritizing high throughput and robustness under heavy load.
- Designed data ingestion workflows across multiple channels, including support tickets, website documentation, and Discord announcements.
- Built the infrastructure using MySQL for chunked data storage and ChromaDB for embeddings.
- Tech Stack:** Python (Flask, Chroma, Celery, Langchain), Redis, MySQL, Docker, Linux

### Machine Learning Engineer

CogniPet AG

April 2019 – December 2019

Zurich, Switzerland (remote)

- Developed a Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance (timed deletion + transparency).
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- Tech Stack:** Python (Flask, Keras, NumPy), AWS (EC2, S3)

### Machine Learning Consultant

Gnanamani Ventures Pvt. Ltd.

May 2019 – August 2019

Namakkal, India (on-site)

- Researched and developed a facial recognition-based attendance system using the FaceNet architecture on NVIDIA Jetson Nano, optimizing for edge computing performance.
- Developed data pipelines to pre-process and fine-tune video data of 30+ staff members at Gnanamani College of Technology, resulting in a successful MVP that automated attendance tracking.
- Tech Stack:** Python (TensorFlow, NumPy), NVIDIA cuDNN, OpenCV, Microsoft Excel

## Projects

### Jix Interpreter | C, CMake, GitHub Actions

- Developed an interpreter for a dynamically typed programming language in C.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.

### SOCKS5 Handler | C++17, CMake, Networking

- Developed a cross-platform C++ library abstracting SOCKS5 proxy communication with async event loop for efficient TCP stream handling, gaining 32+ GitHub stars.
- Implemented RFC 1928/1929 compliant features including remote DNS resolution, IPv4/IPv6 support, and client authentication.

### Prism Coverage | Java, Maven, JUnit, GitHub Actions

- Developed a Java test code coverage engine to measure test suite coverage against control flow and data-flow criteria.
- Implemented color-coded HTML report generation highlighting covered and uncovered sections of source code.
- Built JUnit test suite and GitHub Actions CI/CD workflow to verify coverage engine functionality.

## Technical Skills

**Languages:** Python, Java, JavaScript (ES6), TypeScript, C++ (C++17), C, Bash, HTML5/CSS3  
**Technologies:** Linux, Docker, SQL (MySQL, Postgres), NoSQL (MongoDB, Redis, Neo4j), AWS  
**Frameworks:** Node.js, React.js, Flask, Django, LangChain  
**Tools:** Git, CI/CD (Github Actions, Jenkins), Build tools (CMake, Maven, Vite)